

MCQS

Question and Answers

Mission of Ramsar conservation?

Mission: The Convention's mission is "The conservation and wise use of all wetlands through local and national actions and international cooperation, as a contribution towards achieving sustainable development throughout the world."

Which type of material DNA bank are useful?

Most **DNA** provided by **DNA banks** is used for studies to attempt to develop more productive or more environmentally friendly agricultural **species**. Some **DNA banks** also store the **DNA** of rare or endangered **species** to ensure their survival. The **DNA bank** can be used to compare and analyze **DNA** samples.

What is legislation?

The process of making or enacting laws.

Who is responsible for botanical garden?

Botanical gardens are often run by universities or other scientific research organizations, and often have associated herbaria and research programmers in plant taxonomy or some other aspect of botanical science.

Give different variant of gene variant?

the phenomenon of change in the frequency of alleles (variants of a gene) in a population of organisms due to chance or random events.

Historical Background of CBD?

The **Convention on Biological Diversity**, was opened for signature at the Earth Summit in Rio de Janeiro, Brazil, on June 5, 1992 and by the end of July 1993, 165 countries had signed the treaty. ... The initial sessions were referred to as meetings of the "Ad Hoc Working Group of Experts on **Biological Diversity**."

Name of botanical institute of Pakistan?

• Abdul Wali Khan University Botanical Garden, Mardan • Botanical Garden, Governor's House, Lahore • Botanical Garden, Govt Zamindar College, Gujrat • Danishmandan Botanic Garden, Lahore • Botanical Garden, University of the Punjab, Quaid-e-Azam Campus, Lahore • Faisalabad Botanical Gardens (part of Gatwala Wildlife Park), Faisalabad • Forman Christian College Botanic Garden, Lahore • Government College University Botanic Garden of GCU, Lahore

Effect of climate changes on invertebrate?

Most invertebrates are expected to change their geographical distribution in response to climate change so as to remain in areas to which they are well adapted. This view is strongly supported by sub-fossil evidence of insect distribution during the glaciations and interglacial periods of the Quaternary Period.

Sanctuaries and national parks?

National Parks:

• Hingol National Park • Hazarganji Chiltan National Park • Kirthar National Park • Lal Suhanra National Park • Margalla Hills National Park • Ayubia National Park • Deosai National Park • Chitral Gol National Park • Khunjerab National Park • Machiara National Park.

Wildlife Sanctuaries of Pakistan are; • Astor Wildlife Sanctuary • Baltistan Wildlife Sanctuary • Chasma and Taunsa Barrage Dolphin Sanctuary • Cholistan Wildlife Sanctuary • Hab Dam Wildlife Sanctuary • Kargah Wildlife Sanctuary

Habitat of species that affect climate change?

Global warming resulting from human emissions of greenhouse gases. The consequences include habitat loss; shifts in climatic conditions and in habitats that surpass migrational capabilities; altered competitive relationships.

How plant cope changes climate?

Maintaining insect species that can provide pollination services for a wide range of crops is also vital to the future of agriculture in the face of climate change. Pollinator populations not only need to be able to cope with changing climatic

conditions, they must also be able to provide the pollination services needed to meet increasing demands for food and retain the capacity to adapt to potential changes in the types of crops grown.

Treaty?

“Treaty, a binding formal agreement, contract, or other written instrument that establishes obligations between two or more subjects of international law” • Treaties do not need to follow any special form. • A treaty often takes the form of a contract, but it may be a joint declaration or an exchange of notes.

Strict nature reserve?

Category Ia: Strict Nature Reserve Primary objective To conserve Regionally, Nationally, Globally outstanding ecosystems, Species (occurrences or aggregations), Geo diversity features

Other Objectives To preserve ecosystems, To secure examples of the natural environment for scientific studies, To minimize disturbance through careful planning, To conserve cultural and spiritual values associated with nature.

Distinguishing features The area should generally: • Have a largely complete set of expected native species. • Be capable of being managed to ensure minimal disturbance. • Be free of significant direct intervention by modern humans. • Have a full set of expected native ecosystems, largely intact with intact ecological processes, or processes capable of being restored with minimal management intervention.

Two application of Germplasm conservation?

In fact cryopreservation has been successfully applied for **germplasm conservation**. Plant species e.g rice, wheat, peanut, sugarcane, coconut. ... Any tissue from a plant can be used for cryopreservation e.g. meristems, embryos, endosperm, ovules, seeds, culture plants.

Threatened species of rann kutch?

The site supports many locally and globally **threatened species**, including the great Indian bustard, houbara bustard, sarus crane, and striped hyena and supports more than 1% of the biogeographical population of flamingos.

Impact of climate IGR?

Invertebrates Genetic Resources Invertebrates include a great number of species that perform valuable functions in agro-ecosystems

Threat of FGR?

Forest genetic resources (FGR) are the heritable materials maintained within and among tree and other woody plant species that are of actual or potential economic, environmental, scientific or societal value.

Category three define?

: Category III: National Monument-Feature

Definition: Protected areas set aside to protect a specific natural monument. They are generally quite small protected areas and often have high visitor value.

Primary objective To protect specific outstanding natural features. Their associated biodiversity and habitats

Ramsar sites of Punjab?

There are three Ramsar sites (wetlands of International importance) in the state- Harike, **Kanjli** and Ropar. These wetlands are important habitats for waterfowl, fish and diversity of other flora and fauna including endangered and vulnerable species. Two other wetlands- Ranjit Sagar and Nangal are National wetlands.

When Nagoya protocol adapt?

The protocol was adopted on 29 October 2010 in Nagoya, Japan, and entered into force on 12 October 2014. It has been ratified by 97 parties, which includes 96 UN member states and the European Union. It is the second protocol to the CBD; the first is the 2000 Cartagena Protocol on Biosafety

Issue should not strict nature reserve?

Category Ia: Strict Nature Reserve

Issues for consideration • There are few areas not under some kind of legal or at least traditional ownership, so that finding places that exclude human activity is often problematic. • Most apparent problem is with climate and air pollution • New and emerging diseases. • In an increasingly modified ecology, it may become increasingly difficult to maintain pristine areas through non-intervention.

Application storage technique germplasm?

Germplasm are living genetic resources such as seeds or tissues that are maintained for the purpose of animal and plant breeding, preservation, and other research uses. ... **Germplasm** collections can range from collections of wild species to elite, domesticated breeding lines that have undergone extensive human selection.

What is botanical garden?

It is a garden dedicated to the collection, cultivation and display of a wide range of plants labeled with their botanical names. It may contain specialist plant collections such as tropical plants, or other species of plants.

Long 5 Marks

Social Cluster value of national monument?

Category III areas are likely to hold socio-cultural values as they may have features such as sacred groves, springs, waterfalls, mountains, sea coves etc. of importance to one or more faith groups. These areas are often of significant tourism value and can be managed with the objective of promoting sustainable tourism.

Explain method processing of sampling in DNA Bank? Method processing sample in DNA bank?

Processing of samples: DNA preserved in DNA banks will be stored either within cells and extracted upon retrieval from storage or extracted from cells and purified before storage. The quality of the DNA is expressed through yield, purity, molecular weight, amplification efficiency and authenticity of sequences. The quality of DNA extracted from plant specimens is dependent on the condition of the specimen before storage, the storage environment and the duration of storage. Rapid drying of plant samples with silica gel or lyophilisation helps to preserve the DNA.

Conservation role of botanical garden?

The first **role of botanical gardens** in plant **conservation** is the horticulture and cultivation functions towards the plants. ... With the better growth of plant, our environment also can become better because plants help to absorb the carbon dioxide when undergo photosynthesis process.

Cartagena protocol?

Cartagena protocol was adopted on June 2001 in Cartagena, Spain. It entered into force on September 11th; 2003. Pakistan signed the Cartagena protocol in June 200. Pakistan has ratified it in May 2009

Objectives of Cartagena Protocol on Biosafety

- Contribute to ensuring an adequate level of protection in the field of the safe transfer, handling and use of living modified organisms resulting from modern biotechnology that may have adverse effects on the conservation and sustainable use of biological diversity,
- It takes into account risks to human health, and specifically focusing on transboundary movements.
- It seeks to protect biodiversity from the potential risks of living modified organisms (LMOs) resulting from modern biotechnology.

Why do forest genetics resources matter?

Diversity of **forest genetic resources** enables the potential for a species (or a population) to adapt to climatic changes and related future challenges such as temperature changes, drought, pests, diseases and **forest** fires.

Name lab of plant genetic resources?

Plant Genetic Resources Institute hosts the sole National Genebank of Pakistan for conservation of plant genetic resources and six labs including

1. germplasm exploration lab
2. seed preservation lab
3. in vitro conservation lab
4. germplasm evaluation lab
5. plant introduction and seed health lab
6. Data management lab

Limitation of germplasm conservation?

The **germplasm conservation** through the conventional methods has several **limitations** such as short-lived seeds, seed dormancy, seed-borne diseases, and high inputs of cost and labour. The techniques of **cryo-preservation** (freezing cells and tissues at -196°C) and using cold storages help us to overcome these problems.

The services provided by PEPA?

Pakistan Environmental Protection Act, 1997

- Protection
- Conservation,

- Rehabilitation and
- Improvement of the environment;
- PEPA provide framework for prevention and control of pollution
- Helps in protection of sustainable development.

Long 10 Marks

Impact of Climate changes on FGR? And Climate effect organisms which are changes associate with tree?

Climate change may also result in high variability in temperature and precipitation, with an increase in incidence of extreme events, such as flooding, late frosts and intensive summer droughts, amongst other events. In some areas, such as the Mediterranean and the Neo-tropics, an increase in seasonality is also expected. Under such conditions, natural selection may not result in efficient adaptation because selection pressures are multi-directional, involving traits that may be inversely correlated at the gene level. The standing genetic variation in populations may then not be large enough to create the rare new genotypic combinations that are required. Ecosystems affected by abrupt change may sustain rapid and widespread transformation as ecological tipping points are exceeded. Given the pivotal role of trees in ecosystem function, abrupt climate change impacts on them may thus have profound consequences for forests as a whole. Irreversible loss of ecosystem integrity and function may follow, with replacement by new nonendemic ecosystems.

Effects of changing climate on organisms associated with trees

In particular, changes in the biology of insect pests and diseases may make ecosystems more susceptible to tree mortality. Because of improved environmental conditions for the pest and reduced tree resistance due to increased stress, pests may react to climate change with range expansions and/or increases in attack severity.